

Use-Case Flow Specification

Public Bus Live Tracking & Crowding Estimator — Problem Statement #24

Use Case: UC-02 — View Bus Arrival & Crowding

Primary Actor: Commuter

Includes: UC-03 (Calculate ETA), UC-04 (Estimate Crowding Level)

Preconditions

1. The commuter has the mobile app open and has selected or searched for a valid route or stop (UC-01).
2. At least one bus serving that route/stop is currently transmitting GPS telemetry to the system.

Postconditions

1. The commuter is shown the live ETA and crowding level for every active bus on the selected route/stop.
2. The view remains open and auto-refreshes until the commuter navigates away.

Main Success Scenario

1. Commuter selects a route or stop from the search results (UC-01).
2. System identifies all buses currently active on that route/stop.
3. System invokes UC-03 (Calculate ETA) to obtain the current ETA for each identified bus.
4. System invokes UC-04 (Estimate Crowding Level) to obtain the current crowding label for each bus.
5. System combines bus location, ETA, and crowding level into a single result set.
6. System displays the results to the commuter as a list and on a map, ordered by soonest arrival.
7. System automatically refreshes the displayed ETA and crowding level as new telemetry arrives.
8. Use case ends when the commuter navigates away from the view.

Alternate Flow

6a. No Active Buses Found

- a. System finds no bus currently transmitting telemetry for the selected route/stop.
- b. System displays a "No live buses available" message along with the scheduled timetable, if available.
- c. Commuter may choose another route/stop (return to UC-01) or retry the current one.

6b. Telemetry Feed Interrupted

- a. System detects that a bus's GPS feed has not updated for more than 30 seconds.
- b. System marks that bus's ETA and crowding level as "Last known" with a timestamp instead of removing it.
- c. If the feed does not resume within 2 minutes, the bus is removed from the active list and the commuter is notified.